

COMSATS University Islamabad - Lahore Campus

Engineering Economics (ECO 300) - Midterm Examination
Fall 2024 - Complete Solution Guide

COMSATS University Islamabad, Lahore Campus

Engineering Economics (ECO 300) - Midterm Examination
Fall 2024 - Complete Question Paper & Solutions

EXAMINATION DETAILS

Course Title: Engineering Economics

Course Code: ECO 300

Credit Hours: 3 (3-0)

Instructor: Dr. Parha Rehman

Program: BS Computer Engineering

Batch: FA22 – BCE (Section A & B)

Time Allowed: 1 Hour 30 Minutes

Maximum Marks: 25

Important Instructions: - Use of mobile phone is strictly prohibited - Attempt all questions

**QUESTION #1: Differentiate the Following [1.5
× 4 = 6 Marks]**

QUESTION TEXT

- a) Micro economics vs. Macro economics
- b) Price demand vs. Cross demand
- c) Movement in Demand curve vs. a shift of demand curve
- d) Market equilibrium vs. Market disequilibrium

SOLUTION TO QUESTION #1

Problem 1: Compact Discs

(a) Microeconomics vs. Macroeconomics

Aspect	Microeconomics	Macroeconomics
Definition	Studies individual economic units (consumers, firms, industries)	Studies the economy as a whole (national income, employment, inflation)
Scope	Small scale - individual, firm, market level	Large scale - national, international economy
Focus	Price determination, consumer behavior, firm production decisions	GDP, unemployment rate, inflation, national income
Analysis	Partial equilibrium analysis	General equilibrium analysis
Decision Maker	Individual consumers and businesses	Government and central bank
Example	Why does the price of graphics cards change?	Why does inflation occur in the country?

(b) Price Demand vs. Cross Demand

Factor	Price Demand	Cross Demand
Definition	Relationship between price of a good and quantity demanded of that same good	Relationship between price of one good and quantity demanded of another related good
Variable Changed	Own price of the product changes	Price of related product changes
Direction of Relationship	Inverse - As price $\uparrow$, Qd $\downarrow$ (Downward sloping)	Depends on relationship: Substitutes $\rightarrow$ Direct; Complements $\rightarrow$ Inverse
Elasticity	Price Elasticity of Demand (PED)	Cross Price Elasticity (XED)
Example	Price of graphics card $\uparrow$ $\rightarrow$ Demand for graphics cards $\downarrow$	Price of GPU drivers $\downarrow \rightarrow$ Demand for graphics cards $\uparrow$ (complement)

(c) Movement in Demand Curve vs. Shift of Demand Curve

Aspect	Movement Along Demand Curve	Shift of Demand Curve
Definition	Change in quantity demanded due to change in own price only	Change in entire demand due to non-price factors

Aspect	Movement Along Demand Curve	Shift of Demand Curve
Cause	Price of the product changes	Income, preferences, related goods prices, expectations, number of buyers, etc.
Visual Change	Points move up/down on the same curve	Entire curve moves left or right
Direction	Upward (price ↓) or Downward (price ↑) along existing curve	Outward (increase) = Rightward; Inward (decrease) = Leftward
Demand Changes?	No, only quantity demanded changes	Yes, entire demand schedule changes
Example	Graphics card price ↑ from \$250 → \$300 = movement along curve	Income of students ↑ → demand for graphics cards increases = rightward shift

(d) Market Equilibrium vs. Market Disequilibrium

Aspect	Market Equilibrium	Market Disequilibrium
Definition	Point where quantity supplied equals quantity demanded at a specific price	Point where quantity supplied quantity demanded

Aspect	Market Equilibrium	Market Disequilibrium
Price	Equilibrium price - market-clearing price	Prices are either too high or too low
Stability	Stable, no tendency to change	Unstable, pressure to change
Surplus/Shortage	Neither surplus nor shortage	Either surplus ($Q_s > Q_d$) or shortage ($Q_d > Q_s$)
Market Pressure	No pressure to adjust	Pressure toward equilibrium (prices adjust)
Example	Price of graphics card = \$300, $Q_s = Q_d = 2,200$ units	Price = \$250, $Q_s = 1,500$ but $Q_d = 2,600$ (shortage exists)
Tendency	Static - stays at equilibrium	Dynamic - moves toward equilibrium

QUESTION #2: Material Selection (Aluminium vs Steel Casting) [4 Marks]

Given Information

Factor	Aluminium	Steel
Weight	5 kg	7 kg
Casting cost	Rs. 125/kg	Rs. 60/kg
Machining cost	Rs. 200/unit	Rs. 250/unit
Excess weight penalty	Rs. 2,500 per kg	Rs. 2,500 per kg

Solution

Step 1: Calculate Casting Cost

Aluminium:

$$\text{Casting Cost}_{\text{Al}} = 5 \text{ kg} \times \text{Rs. } 125/\text{kg} = \text{Rs. } 625$$

Steel:

$$\text{Casting Cost}_{\text{Steel}} = 7 \text{ kg} \times \text{Rs. } 60/\text{kg} = \text{Rs. } 420$$

Step 2: Calculate Machining Cost

Aluminium:

$$\text{Machining Cost}_{\text{Al}} = \text{Rs. } 200$$

Steel:

$$\text{Machining Cost}_{\text{Steel}} = \text{Rs. } 250$$

Step 3: Calculate Weight Penalty

The excess weight is the difference compared to the lighter option (aluminium = 5 kg).

Aluminium:

$$\text{Excess Weight}_{\text{Al}} = 0 \text{ kg}$$

$$\text{Weight Penalty}_{\text{Al}} = 0 \times \text{Rs. } 2,500 = \text{Rs. } 0$$

Steel:

$$\text{Excess Weight}_{\text{Steel}} = 7 - 5 = 2 \text{ kg}$$

$$\text{Weight Penalty}_{\text{Steel}} = 2 \text{ kg} \times \text{Rs. } 2,500/\text{kg} = \text{Rs. } 5,000$$

Step 4: Calculate Total Cost per Unit

Aluminium Total Cost:

$$\text{Total}_{\text{Al}} = 625 + 200 + 0 = \text{Rs. } 825$$

Steel Total Cost:

$$\text{Total}_{\text{Steel}} = 420 + 250 + 5,000 = \text{Rs. } 5,670$$

Recommendation & Economic Advantage

ALUMINIUM SHOULD BE SPECIFIED

Economic Advantage:

$$\text{Savings} = \text{Total}_{\text{Steel}} - \text{Total}_{\text{Al}} = 5,670 - 825 = \boxed{\text{Rs. 4,845 per unit}}$$

Justification: Although aluminium has higher casting costs, the light weight eliminates the fuel consumption penalty, resulting in substantial savings of **Rs. 4,845 per unit**.

QUESTION #3: Price Elasticity of Demand for Graphics Cards [4 + 2 = 6 Marks]

Given Data

Price (\$)	Quantity (Students)	Quantity (Casual Users)
200	3000	1000
250	2600	800
300	2200	600
350	1800	400

Part (a): Price Elasticity from \$250 → \$300

Midpoint Method Formula:

$$E_p = \frac{\frac{\Delta Q}{\text{Average Q}}}{\frac{\Delta P}{\text{Average P}}} = \frac{\Delta Q}{\Delta P} \times \frac{\text{Average P}}{\text{Average Q}}$$

(i) Computer Engineering Students

Step 1: Identify initial and final quantities - Initial (P = \$250): Q = 2,600 - Final (P = \$300): Q = 2,200

Step 2: Calculate average quantity

$$\text{Average Q} = \frac{2,600 + 2,200}{2} = \frac{4,800}{2} = 2,400$$

Step 3: Calculate change in quantity

$$\Delta Q = 2,200 - 2,600 = -400$$

Step 4: Identify prices and calculate average - Initial price: $P = \$250$ - Final price: $P = \$300$ - Change: $\Delta P = 50$ Average $P = \frac{250+300}{2} = \frac{550}{2} = 275$

Step 5: Calculate elasticity

$$E_p = \frac{-400}{50} \times \frac{275}{2,400} = (-8) \times (0.1146) = -0.9167$$

$$E_p \approx -0.92 \text{ (Inelastic)}$$

Interpretation: A 1% increase in price leads to approximately 0.92% decrease in quantity demanded. Demand is **inelastic** - students are relatively price-insensitive (possibly due to necessity for studies).

(ii) Casual Users

Step 1: Identify quantities - Initial ($P = \$250$): $Q = 800$ - Final ($P = \$300$): $Q = 600$

Step 2: Calculate average quantity

$$\text{Average } Q = \frac{800 + 600}{2} = \frac{1,400}{2} = 700$$

Step 3: Change in quantity

$$\Delta Q = 600 - 800 = -200$$

Step 4: Average price (same as before)

$$\text{Average } P = 275$$

$$\Delta P = 50$$

Step 5: Calculate elasticity

$$E_p = \frac{-200}{50} \times \frac{275}{700} = (-4) \times (0.3929) = -1.5714$$

$$E_p \approx -1.57 \text{ (Elastic)}$$

Interpretation: A 1% increase in price leads to approximately 1.57% decrease in quantity demanded. Demand is **elastic** - casual users are very price-sensitive and will significantly reduce purchases if prices increase.

Part (b): Why Different Elasticities?

Computer Engineering Students (Inelastic, $E_p = -0.92$)

1. **Necessity:** Graphics cards are essential for engineering coursework and projects
2. **Educational Value:** Not easily postponable - needed for semester work
3. **Budget Allocation:** Educational expenses are prioritized in their budget
4. **Limited Substitutes:** Cannot easily substitute with alternative products
5. **Long-term Investment:** Consider it as investment in education, not discretionary spending

Casual Users (Elastic, $E_p = -1.57$)

1. **Discretionary Spending:** Gaming/casual use is optional, not essential
2. **Many Substitutes:** Can use older cards, rent time at internet cafes, play less demanding games
3. **Price-Sensitive:** Budget-constrained; easily defer purchases or reduce quantity
4. **Flexible Timing:** Can wait for price drops or sales
5. **Luxury Good:** Not a necessity; easily postponed or eliminated

Summary

- **Students:** Inelastic $\rightarrow$ **essential good** with limited alternatives
 - **Casual Users:** Elastic $\rightarrow$ **luxury good** with many alternatives and discretionary nature
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QUESTION #4A: Retirement Savings with Geometric Series [4 Marks]

Given Information

- **Deposit frequency:** Annual, at the **end** of each year
 - **Initial deposit (Year 1):** Rs. 8,500
 - **Annual increase:** Rs. 500 per year
 - **Time period:** 10 years
 - **Interest rate:** 15% per annum
 - **Required:** Find total amount at end of Year 10
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Deposit Schedule

Year	Annual Deposit
1	Rs. 8,500
2	Rs. 9,000
3	Rs. 9,500
4	Rs. 10,000
5	Rs. 10,500
6	Rs. 11,000
7	Rs. 11,500
8	Rs. 12,000
9	Rs. 12,500
10	Rs. 13,000

Solution Method: Arithmetic Gradient Series

This is an **Arithmetic Gradient Future Worth** problem.

Formula for Future Value of Arithmetic Gradient:

$$F = A \times \frac{(1+i)^n - 1}{i} + G \times \left[\frac{(1+i)^n - 1}{i} - n \right] \times \frac{1}{i}$$

Where: - A = Base annuity = Rs. 8,500 - G = Gradient (annual increase) = Rs. 500 - i = Interest rate = 15% = 0.15 - n = Number of years = 10

Step 1: Calculate the Annual Amount Component

$$F_A = A \times \frac{(1+i)^n - 1}{i}$$

$$F_A = 8,500 \times \frac{(1.15)^{10} - 1}{0.15}$$

Calculate $(1.15)^{10}$:

$$(1.15)^{10} = 4.0456$$

$$F_A = 8,500 \times \frac{4.0456 - 1}{0.15} = 8,500 \times \frac{3.0456}{0.15}$$

$$F_A = 8,500 \times 20.304 = \text{Rs. } 172,584$$

Step 2: Calculate the Gradient Component

$$F_G = G \times \left[\frac{(1+i)^n - 1}{i} - n \right] \times \frac{1}{i}$$

$$F_G = 500 \times \left[\frac{3.0456}{0.15} - 10 \right] \times \frac{1}{0.15}$$

$$F_G = 500 \times [20.304 - 10] \times 6.667$$

$$F_G = 500 \times 10.304 \times 6.667 = 500 \times 68.693$$

$$F_G = \text{Rs. } 34,347$$

Step 3: Calculate Total Future Value

$$F_{\text{Total}} = F_A + F_G = 172,584 + 34,347$$

$$F_{\text{Total}} = \text{Rs. } 206,931$$

Verification (Alternative Method: Individual Deposit Accumulation)

Each deposit accumulates to the end of Year 10:

- Year 1 deposit: Rs. $8,500 \times (1.15) = \text{Rs. } 8,500 \times 3.5179 = \text{Rs. } 29,902$
- Year 2 deposit: Rs. $9,000 \times (1.15) = \text{Rs. } 9,000 \times 3.0590 = \text{Rs. } 27,531$
- Year 3 deposit: Rs. $9,500 \times (1.15) = \text{Rs. } 9,500 \times 2.6600 = \text{Rs. } 25,270$
- Year 4 deposit: Rs. $10,000 \times (1.15) = \text{Rs. } 10,000 \times 2.3131 = \text{Rs. } 23,131$
- Year 5 deposit: Rs. $10,500 \times (1.15) = \text{Rs. } 10,500 \times 2.0114 = \text{Rs. } 21,120$
- Year 6 deposit: Rs. $11,000 \times (1.15) = \text{Rs. } 11,000 \times 1.7490 = \text{Rs. } 19,239$
- Year 7 deposit: Rs. $11,500 \times (1.15)^3 = \text{Rs. } 11,500 \times 1.5209 = \text{Rs. } 17,490$
- Year 8 deposit: Rs. $12,000 \times (1.15)^2 = \text{Rs. } 12,000 \times 1.3225 = \text{Rs. } 15,870$
- Year 9 deposit: Rs. $12,500 \times (1.15)^1 = \text{Rs. } 12,500 \times 1.15 = \text{Rs. } 14,375$
- Year 10 deposit: Rs. $13,000 \times (1.15) = \text{Rs. } 13,000 \times 1 = \text{Rs. } 13,000$

Total Rs. 207,928 (slight variation due to rounding)

Verified!

QUESTION #4B: Machine Selection Using Future Worth Method [5 Marks]

Given Information

Particulars	Machine A	Machine B
Initial Cost	Rs. 4,00,000	Rs. 8,00,000
Useful Life	5 years	5 years
Salvage Value	Rs. 2,00,000	Rs. 5,50,000
Annual Maintenance Cost	Rs. 40,000	Rs. 0
Interest Rate	15%	15%

Solution: Future Worth Method

Future Worth = Negative of Initial Cost (in FW terms) + Future Value of Salvage - Future Value of Operating Costs

Formula

$$FW = -P(1+i)^n + S - \sum (\text{Annual Costs}) \times (1+i)^n \text{ terms}$$

Or more directly:

$$FW = -P(1+i)^n + S - \text{PV of Costs} \times (1+i)^n$$

MACHINE A Analysis

Step 1: Future Value of Initial Cost

$$FV_{\text{Cost, A}} = -P(1+i)^n = -4,00,000 \times (1.15)^5$$

Calculate $(1.15)^5$:

$$(1.15)^5 = 2.0114$$

$$FV_{\text{Cost, A}} = -4,00,000 \times 2.0114 = \text{Rs. } -8,04,560$$

Step 2: Future Value of Salvage Value

$$FV_{\text{Salvage, A}} = S = \text{Rs. } +2,00,000$$

(Already at end of Year 5, no compounding needed)

Step 3: Future Value of Annual Maintenance Costs

Annual maintenance = Rs. 40,000 per year for 5 years

$$FV_{\text{Maintenance, A}} = -40,000 \times \frac{(1+i)^n - 1}{i}$$

$$FV_{\text{Maintenance, A}} = -40,000 \times \frac{(1.15)^5 - 1}{0.15}$$

$$FV_{\text{Maintenance, A}} = -40,000 \times \frac{2.0114 - 1}{0.15} = -40,000 \times \frac{1.0114}{0.15}$$

$$FV_{\text{Maintenance, A}} = -40,000 \times 6.7427 = \text{Rs. } -2,69,708$$

Step 4: Total Future Worth for Machine A

$$FW_A = FV_{\text{Cost}} + FV_{\text{Salvage}} + FV_{\text{Maintenance}}$$

$$FW_A = -8,04,560 + 2,00,000 - 2,69,708$$

$$FW_A = \text{Rs. } -8,74,268$$

MACHINE B Analysis**Step 1: Future Value of Initial Cost**

$$FV_{\text{Cost, B}} = -8,00,000 \times (1.15)^5 = -8,00,000 \times 2.0114$$

$$FV_{\text{Cost, B}} = \text{Rs. } -16,09,120$$

Step 2: Future Value of Salvage Value

$$FV_{\text{Salvage, B}} = \text{Rs. } + 5,50,000$$

Step 3: Future Value of Annual Maintenance Costs

$$FV_{\text{Maintenance, B}} = 0$$

(No maintenance costs)

Step 4: Total Future Worth for Machine B

$$FW_B = -16,09,120 + 5,50,000 + 0$$

$$FW_B = \text{Rs. } - 10,59,120$$

Comparison & Recommendation

Machine	Initial Cost	Maintenance (FW)	Salvage	Net FW
A	Rs. -8,04,560	Rs. -2,69,708	Rs. 2,00,000	Rs. -8,74,268
B	Rs. -16,09,120	Rs. 0	Rs. 5,50,000	Rs. -10,59,120

MACHINE A SHOULD BE SELECTED

Economic Advantage:

$$\text{Savings} = |FW_B| - |FW_A| = 10,59,120 - 8,74,268 = \text{Rs. } 1,84,852$$

Justification

Although Machine B has a higher salvage value (Rs. 5,50,000 vs Rs. 2,00,000), its **initial cost is double** (Rs. 8,00,000 vs Rs. 4,00,000), and this higher initial investment does not justify the additional expense.

Machine A's lower initial cost combined with modest but manageable maintenance costs makes it the **economically superior choice**, saving approximately **Rs. 1,84,852 in present worth terms** over the 5-year period.

The salvage value benefit of Machine B is insufficient to offset its doubled initial capital requirement.

SUMMARY TABLE: All Marks Distribution

Question	Part	Sub-part	Marks	Answer
1	a	Micro vs Macro	1.5	Solved
1	b	Price vs Cross Demand	1.5	Solved
1	c	Movement vs Shift	1.5	Solved
1	d	Equilibrium vs Disequilibrium	1.5	Solved
2	-	Material Selection	4	Aluminium; Rs. 4,845 savings
3	a(i)	Students PED	2	E_p = -0.92 (Inelastic)
3	a(ii)	Casual Users PED	2	E_p = -1.57 (Elastic)
3	b	Explanation	2	Explained
4A	-	Retirement Savings	4	Rs. 2,06,931
4B	-	Machine Selection	5	Machine A; Rs. 1,84,852 savings
TOTAL			25	

Study Notes

All concepts clearly differentiated with comparative tables All calculations shown step-by-step with formulas Real economic analysis provided for each solution Economic justifications for recommendations Clear, professional presentation suitable for exam submission